

RecallGuard architecture

Revision: 2026-09-26. Implemented local core.

Request boundary

Browser playground or API client -> Flask validation -> API-key authentication -> owner identity.

Policy boundary

MemoryService -> consent and sensitive-pattern admission -> exact duplicate check -> shared normalized embedding encoder -> owner-scoped MemoryStore -> Chroma persistent collection.

Retrieval boundary

Owner-scoped active IDs -> semantic candidates -> similarity or exact-term coverage -> confidence/type filters -> multi-signal ranking -> bounded serialized context -> local Ollama.

Maintenance boundary

Separate worker/one-shot -> per-owner expiry purge -> duplicate consolidation -> low-confidence cleanup -> content-free summary.

Test boundary

App factory and injectable dependencies -> temporary Chroma databases -> deterministic or real embedding encoder -> mock or real Ollama. Existing user storage is never selected by test runners.

Core invariants

Owner identity cannot come from request JSON. Every object operation requires owner identity. Expired records do not enter candidates. Privacy admission runs before embeddings. Correction replaces old content. Deletion removes source record/vector. Context reserves instructions/envelope and output budget.

Deferred public architecture

Render, Supabase and Groq are selected future providers, not implemented adapters in this revision. The container package runs the local stack with a durable volume and host Ollama; it is not claimed to fit Render free memory.